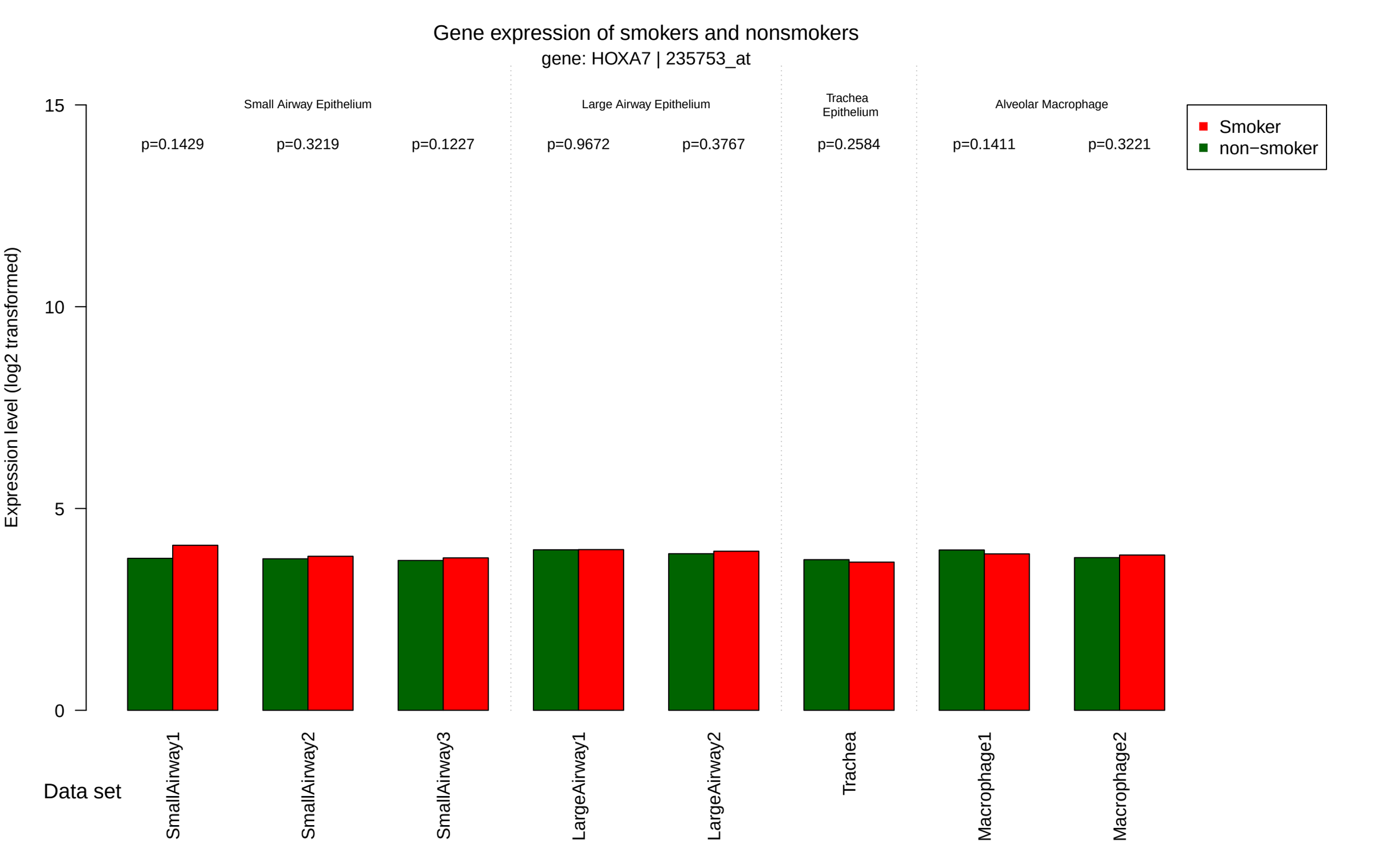
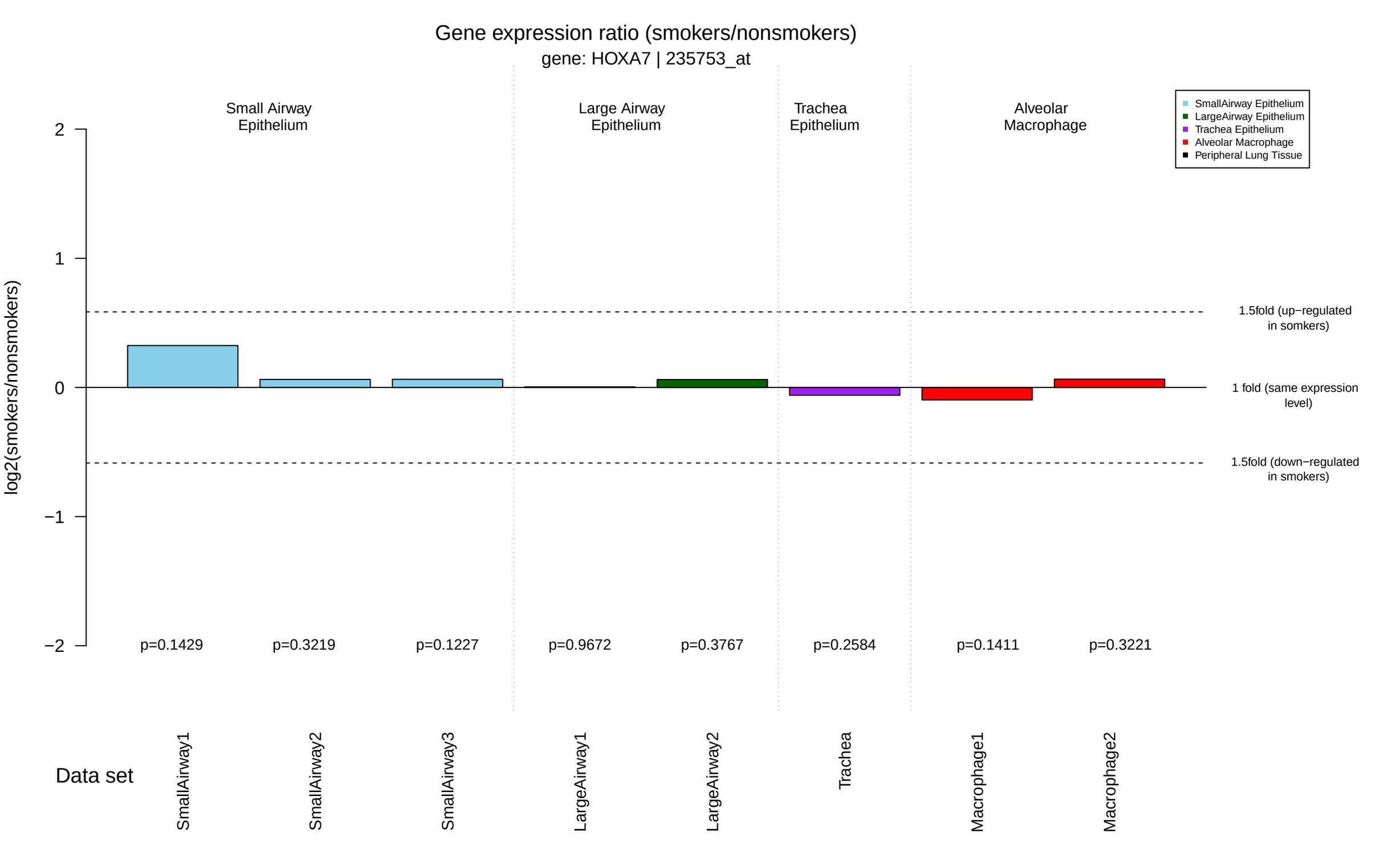
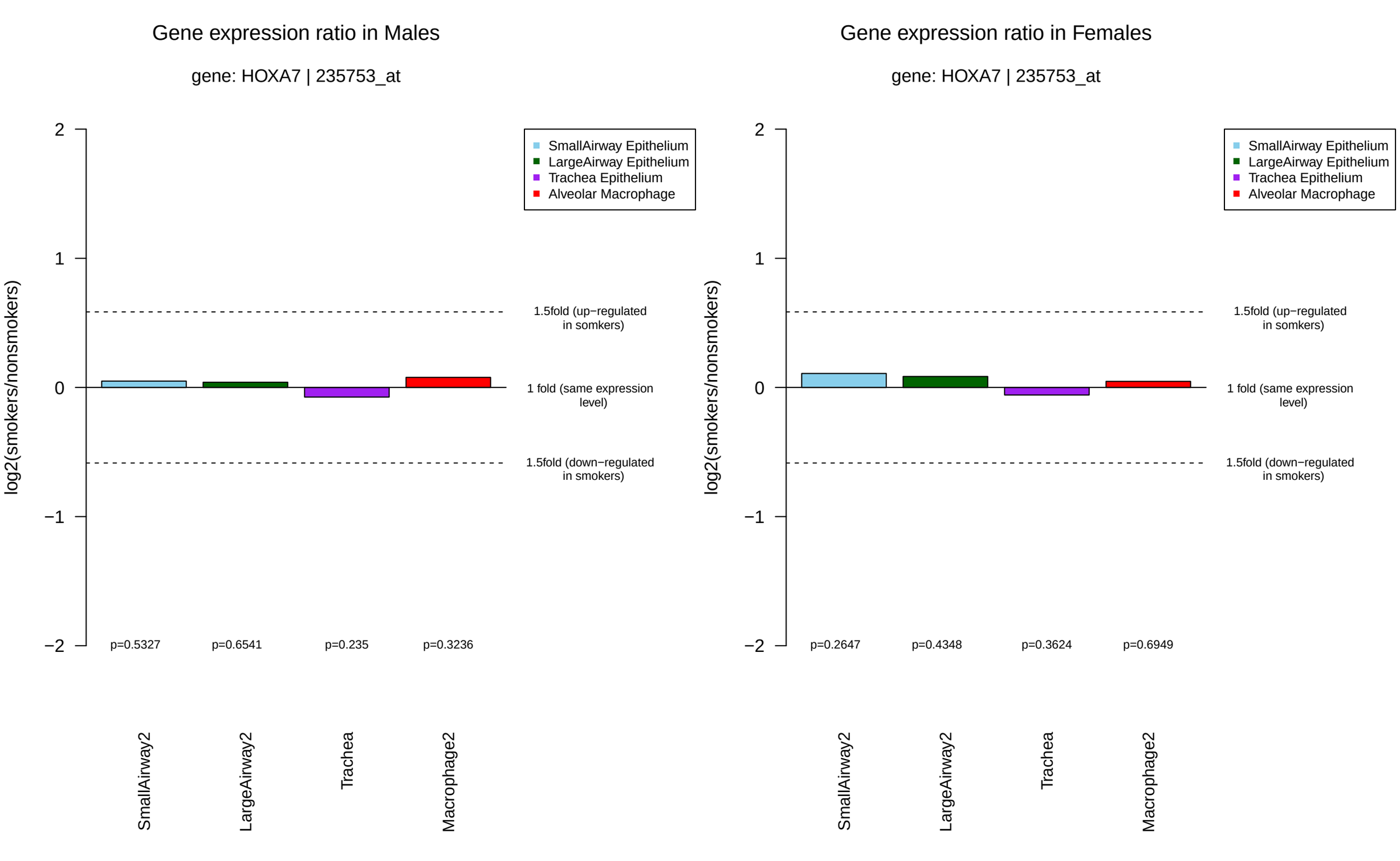
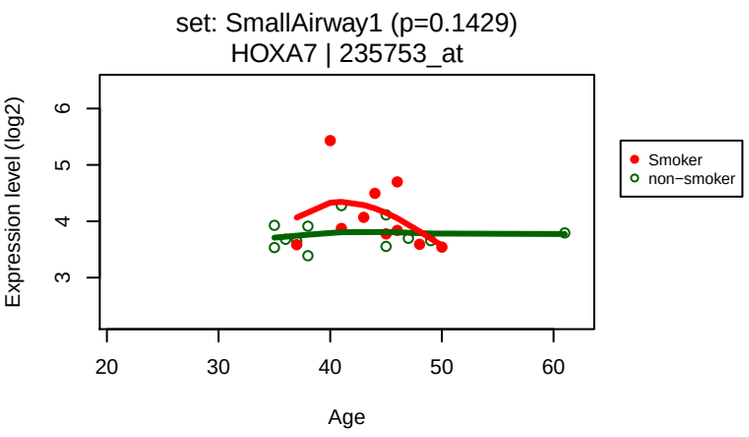
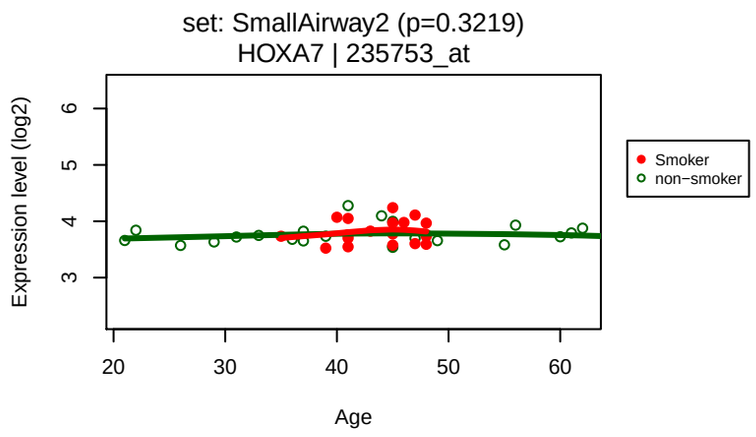
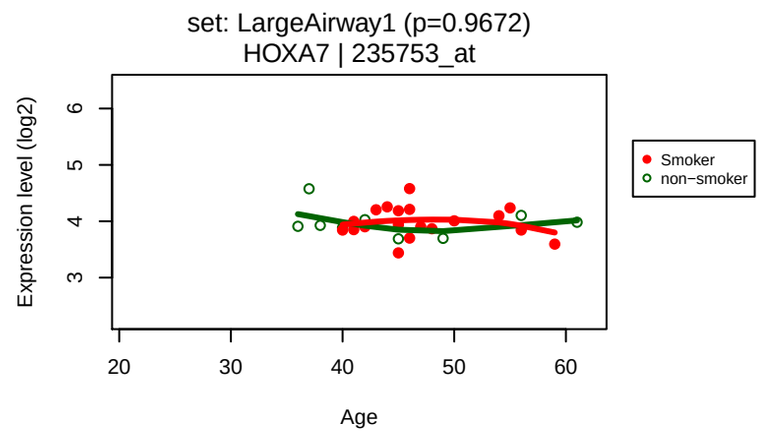

Gene expression of smokers and nonsmokers
gene: HOXA7 | 235753_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1429
p=0.3219
p=0.1227
p=0.9672
p=0.3767
p=0.2584
p=0.1411
p=0.3221
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

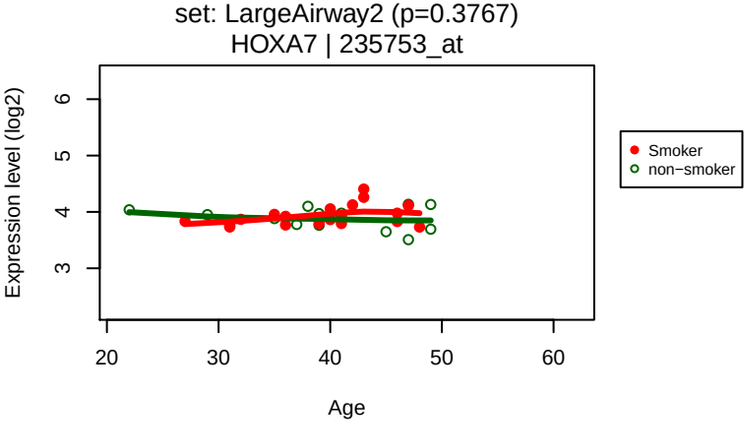

Gene expression ratio (smokers/nonsmokers)
gene: HOXA7 | 235753_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1429
p=0.3219
p=0.1227
p=0.9672
p=0.3767
p=0.2584
p=0.1411
p=0.3221
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

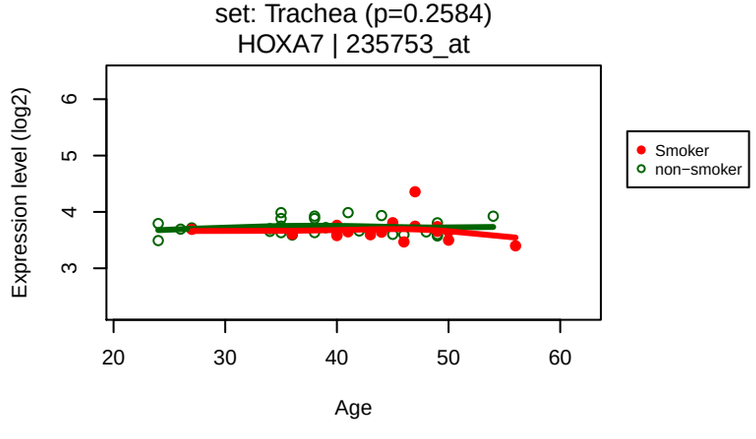

Gene expression ratio in Males
gene: HOXA7 | 235753_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5327
p=0.6541
p=0.235
p=0.3236
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOXA7 | 235753_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2647
p=0.4348
p=0.3624
p=0.6949
SmallAirway2
LargeAirway2
Trachea
Macrophage2

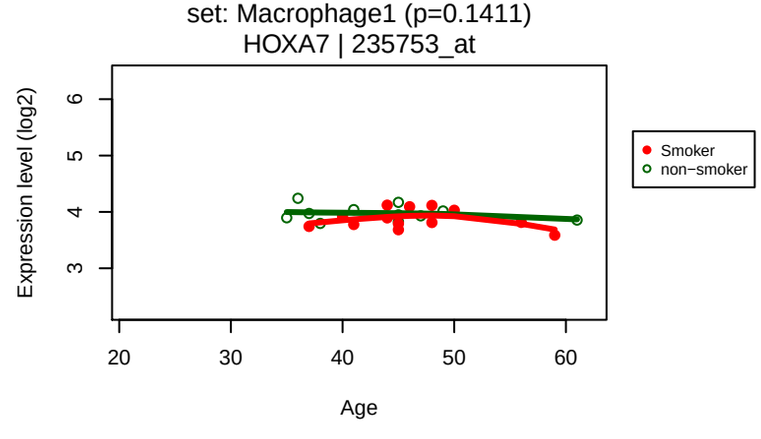

set: SmallAirway1 (p=0.1429)
HOXA7 | 235753_at
Expression level (log2)
Age
Smoker
non−smoker

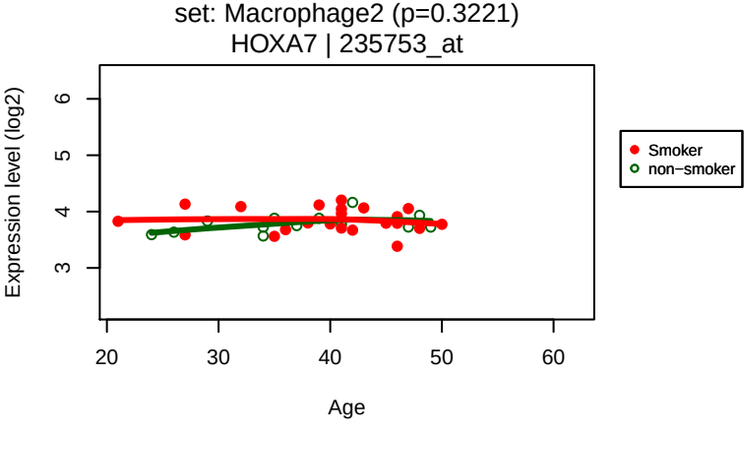

set: SmallAirway2 (p=0.3219)
HOXA7 | 235753_at
Expression level (log2)
Age
Smoker
non−smoker

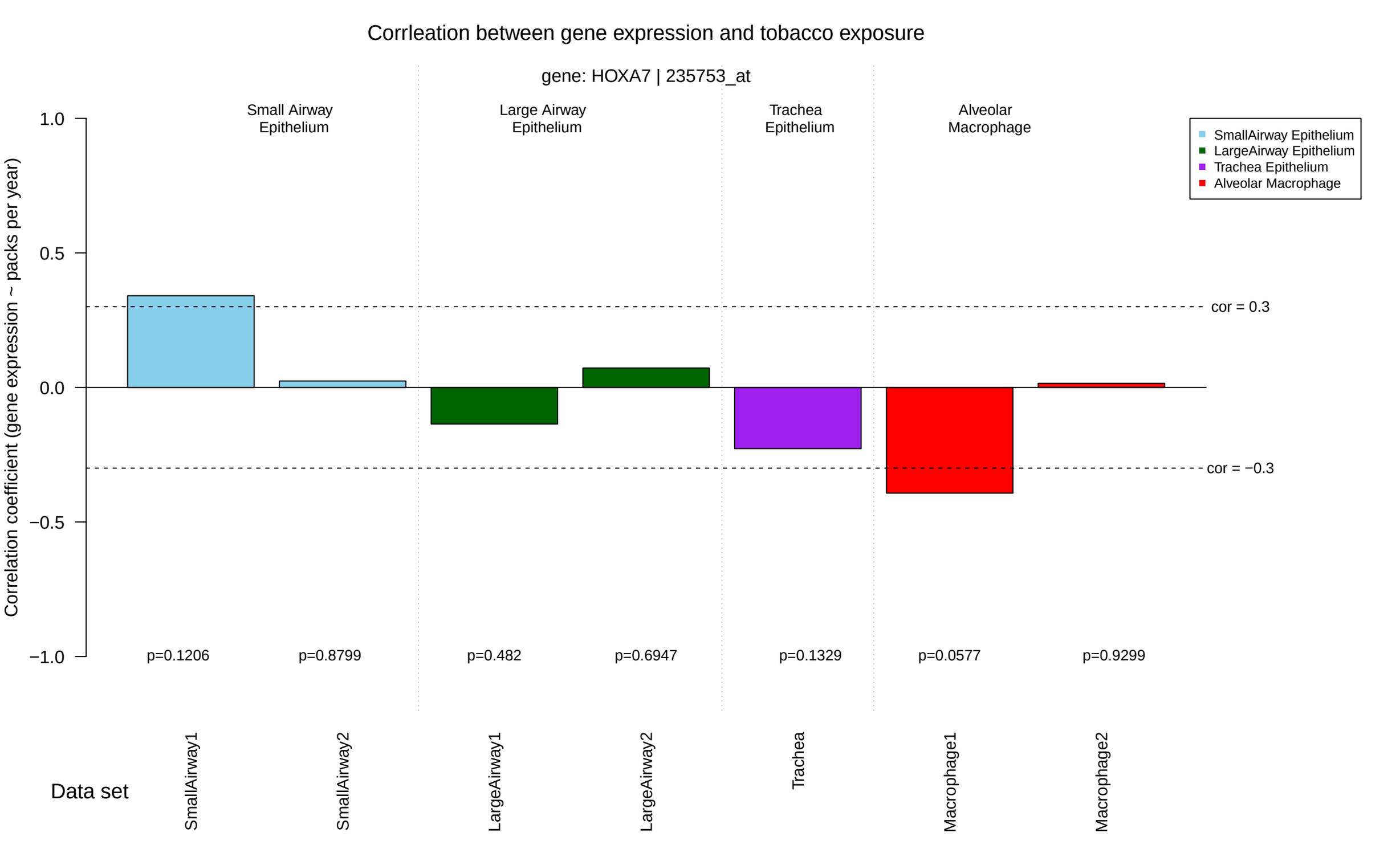

set: LargeAirway1 (p=0.9672)
HOXA7 | 235753_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3767)
HOXA7 | 235753_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2584)
HOXA7 | 235753_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1411)
HOXA7 | 235753_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3221)
HOXA7 | 235753_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOXA7 | 235753_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1206
p=0.8799
p=0.482
p=0.6947
p=0.1329
p=0.0577
p=0.9299
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2